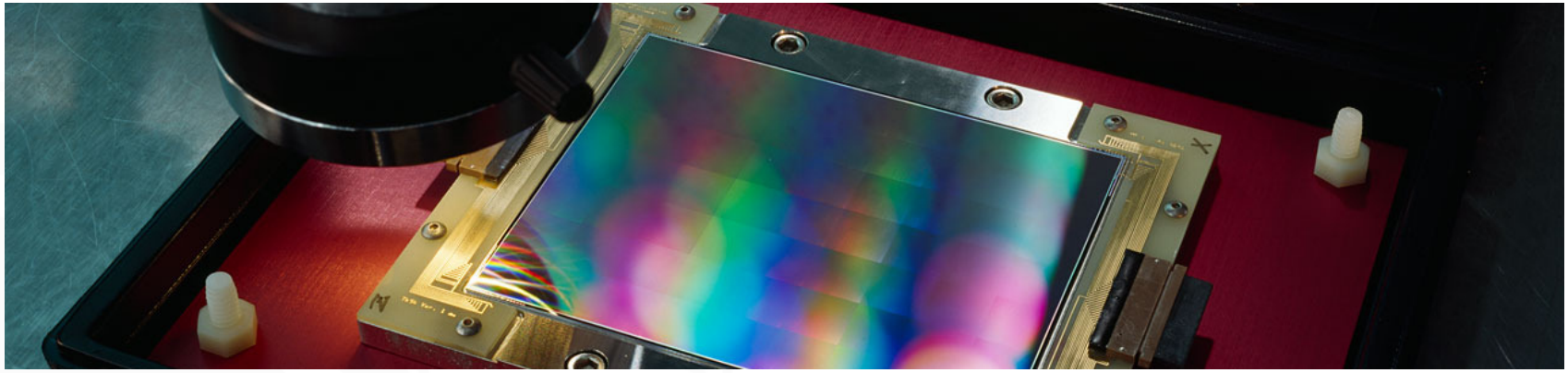


Introduction to CCD detectors for astronomy

M1 Astro-CCP HAP 703P

Atelier Astrophysique Observationnelle 1

Julien Morin

23rd September 2026

UNIVERSITÉ MONTPELLIER

FACULTÉ DES SCIENCES DE MONTPELLIER

LUPM LABORATOIRE MONTPELLIER UNIVERS & PARTICULES

Outline

1. Enseignement d'astrophysique en Master Astrophysique/CCP
2. The CCD revolution in astronomy
3. Structure and operation of a CCD
4. Defects of CCD detectors and frame processing
5. CCD Cameras available at UM
6. Bibliographic ressources

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1 Enseignement d'astrophysique en Master Astrophysique/CCP

2 The CCD revolution in astronomy

3 Structure and operation of a CCD

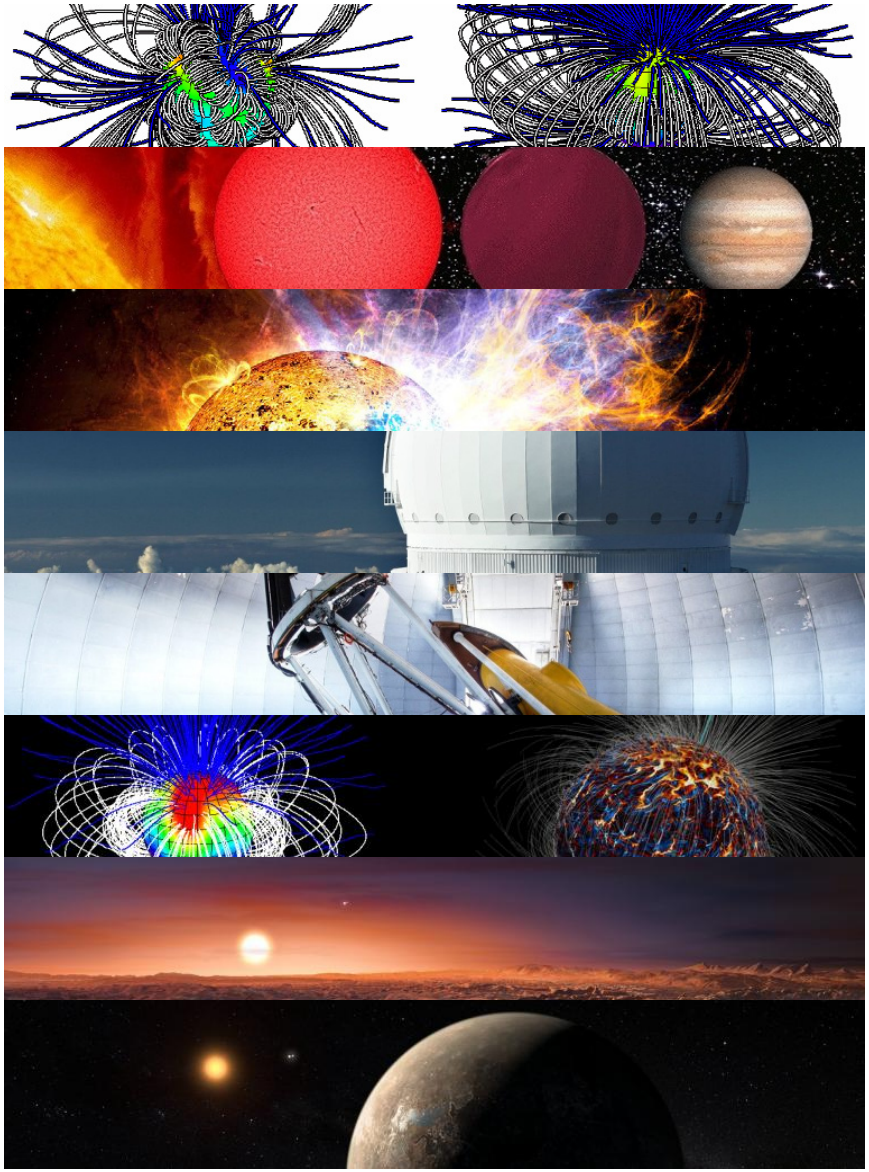
4 Defects of CCD detectors and frame processing

5 CCD Cameras available at UM

6 Bibliographic ressources

Présentations

- Julien Morin
  - julien.morin@umontpellier.fr
- Maître de Conférences
  - Faculté des Sciences
    - Physique
    - Astrophysique
  - Laboratoire Univers et Particules de Montpellier
    - Magnétisme stellaire
    - Planètes extrasolaires



Enseignement d'astrophysique en Master Astrophysique/CCP

- M1 AstroCCP
  - HAP 702P Astrophysique
  - HAP 703P Astro. Obs. 1
- M2 Astrophysique et CCP
  - Transfert de rayonnement et atmosphères stellaires
  - Milieu interstellaire
  - Astro. Obs. 2
- M2 Astrophysique
  - Structure et évolution stellaires
  - Projet Num. Astro.
  - Form. Etoiles Syst. Planét.
  - Form. Evol. Galaxies
  - Cosmologie observationnelle
  - Instru. Astrophysique

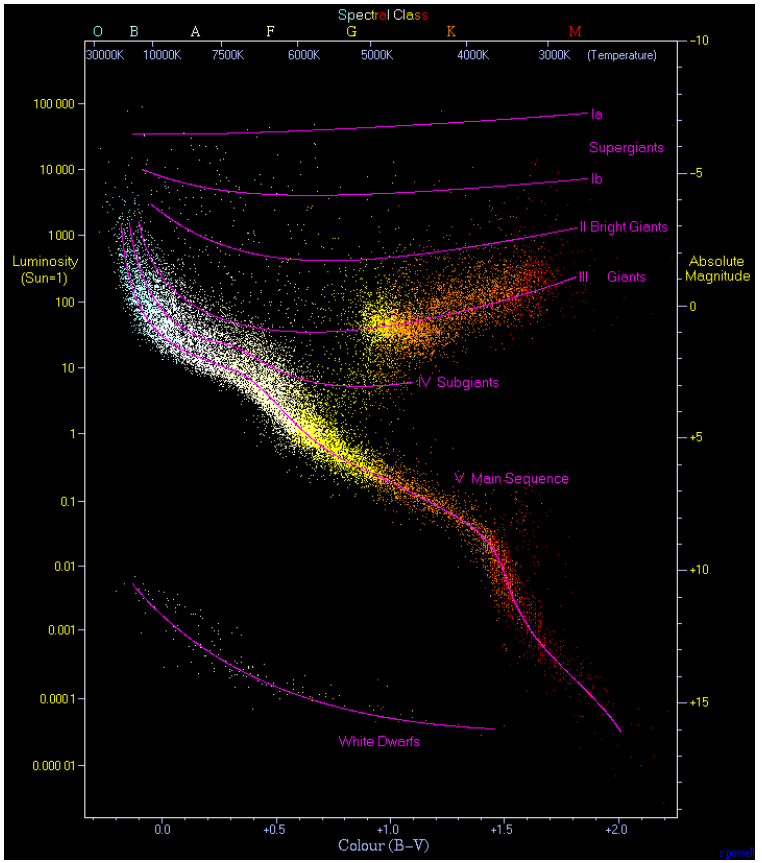
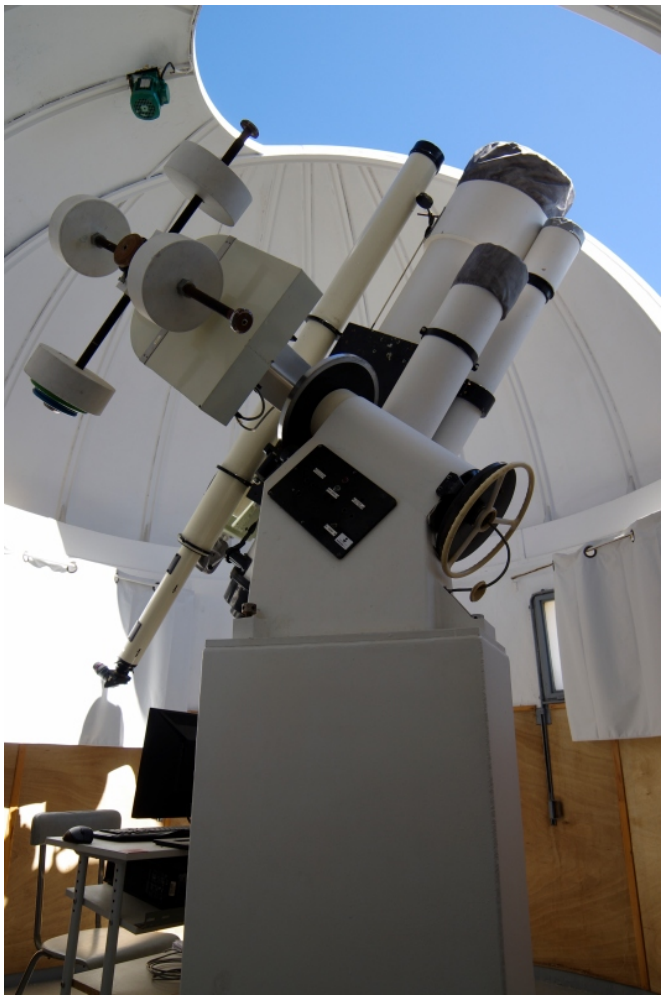


Diagramme de Hertzsprung-Russel

Crédit: R. Powell

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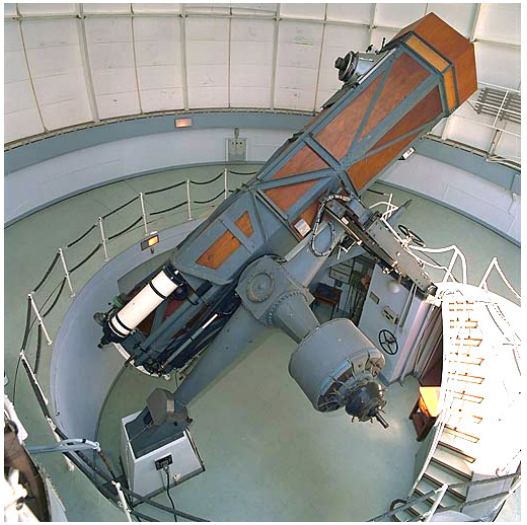


Coupole Est de l'observatoire universitaire

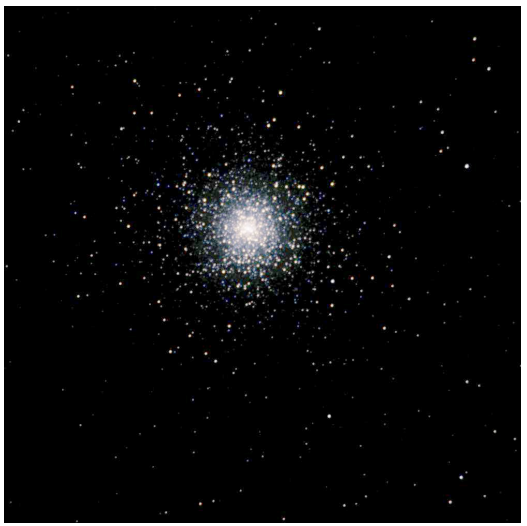
Crédit: J. Morin

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OHP/T120 Crédit : Ministère de la Culture



M15 Crédit : D. Minenna/V. Niederkorn/R. Ruffault

Déroulement M1 HAP 702P/703P

- HAP 702P : B. Plez ; 24h CM+TD ; vendredi a-m ; 1er cours : 25/09
  - Astronomie de position
  - Processus de rayonnement et photométrie
  - Spectroscopie astrophysique
  - Photosphères stellaires transfert de rayonnement
- HAP 703P : cours/TP (21 h) + TP Obs (3 h) + TP autonome
  - 23/09 : Utilisation des détecteurs CCD en astronomie
  - 30/09 : Photométrie astrophysique
  - 14/10 : Spectroscopie astrophysique + Présentation des projets
  - 21/10 : Modélisation des observations
- `https://moodle.umontpellier.fr/`
  - cours HAP 703P
  - observatoire FdS

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3 Structure and operation of a CCD

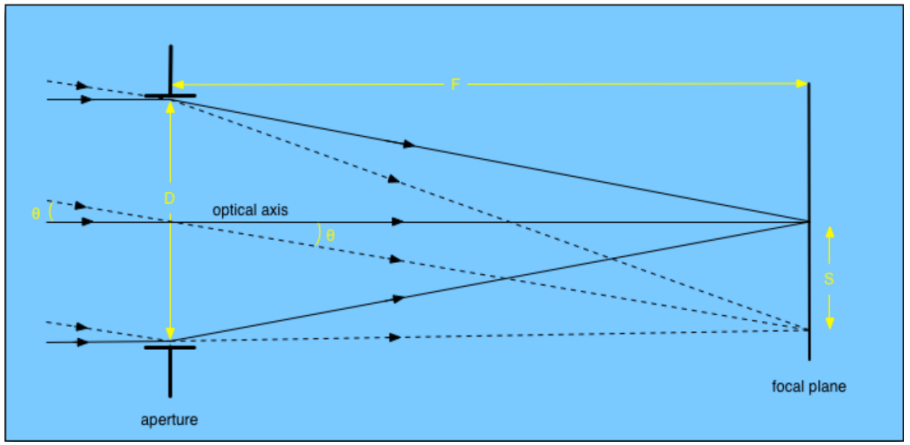
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The CCD revolution in astronomy (1/2)

- The detector records the image formed in the focal plane of the telescope
- Previous detectors
  - Eye
  - Photographic plate
  - Photomultiplier (PMT)
  - Television camera
  - Image intensifier
- Charge-couple device (CCD)
  - Boyle & Smith (1969)
  - Nobel prize in Physics 2009
  - Si semi-conductor capacitors

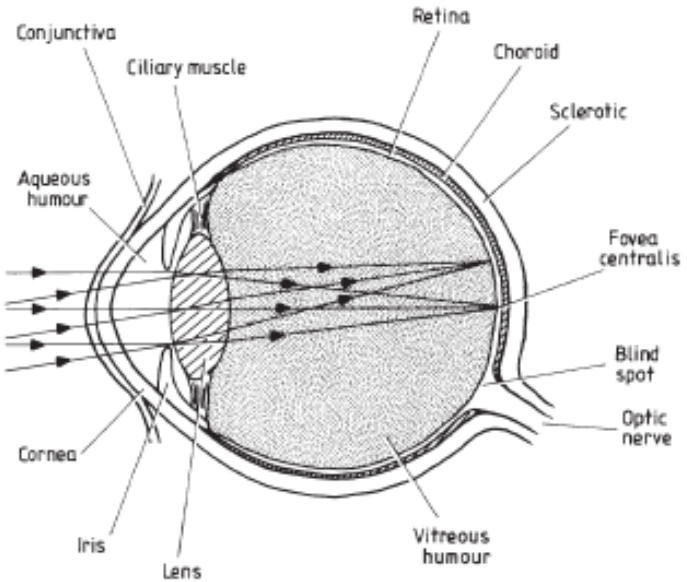


Optical layout of a refractor telescope

Credit: S. Littlefair

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Optical layout of the human eye

Credit: R. Kitchin, Astrophysical Techniques

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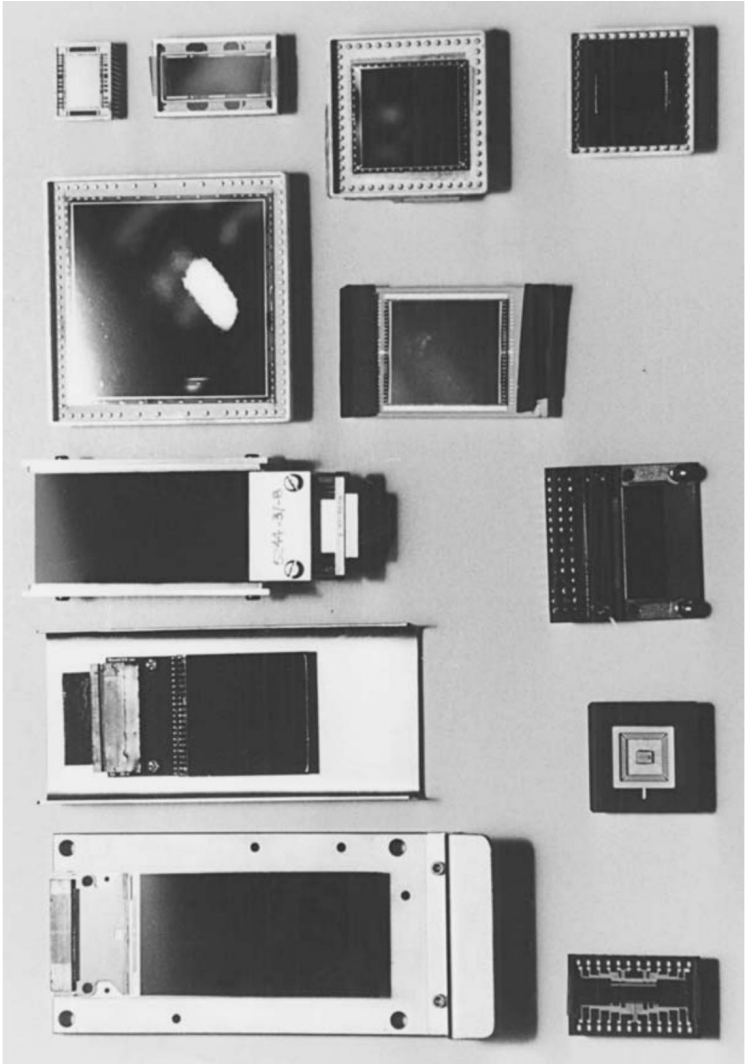


Photographic plate image of galaxy M51 (1994)

Credit: Mount Palomar 48" Schmidt Telescope / ESO Online Digitized Sky Survey (DSS2-red)

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Examples of CCD detector chips

Credit: S.B. Howell, Handbook of CCD astronomy

The CCD revolution in astronomy (2/2)

- Advantages over other technologies
  - Quantum efficiency
    - $QE = \frac{n_{\gamma,\mathrm{det}}}{n_\gamma}$
    - peak $> 90\%$
  - Bandpass
    - $300 - 1\,100$ nm
  - Linearity
  - Dynamic range
    - $> 10^4$
  - Noise properties
  - Fast readout, 2-D arrays,...



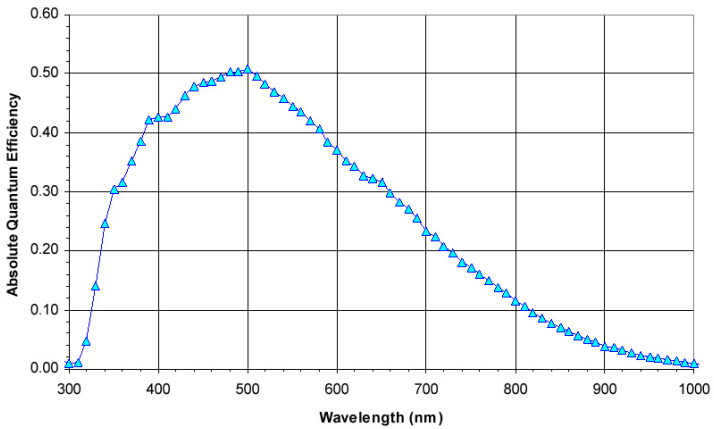


Figure 11 - Monochrome with Microlens Quantum Efficiency

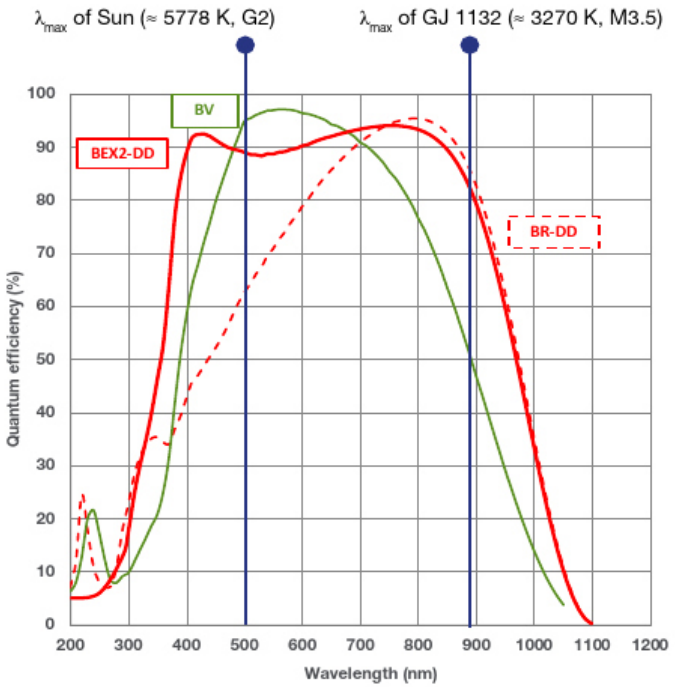
SBIG STL-11000M and QE curve

Credit: Santa Barbara Instrument Group

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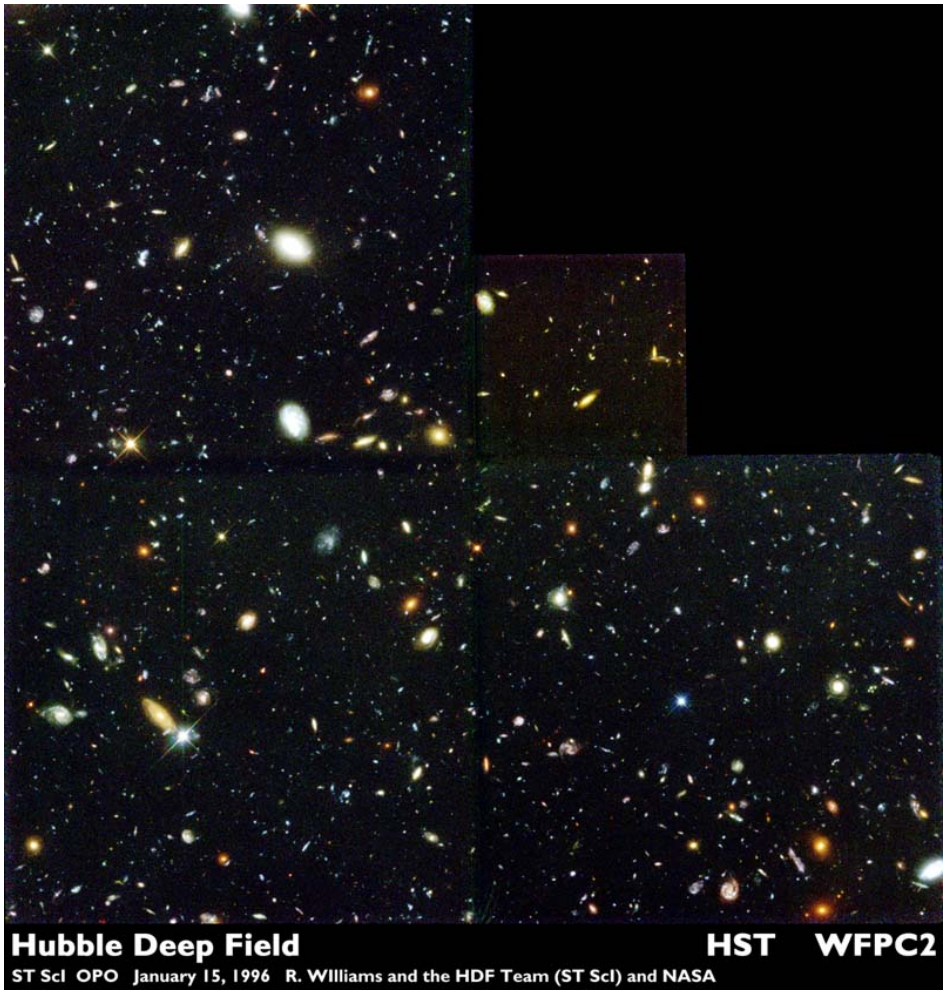


ANDOR iKon-L 936 and QE curve

Credit: Oxford Instruments

The CCD revolution in astronomy (2/2)

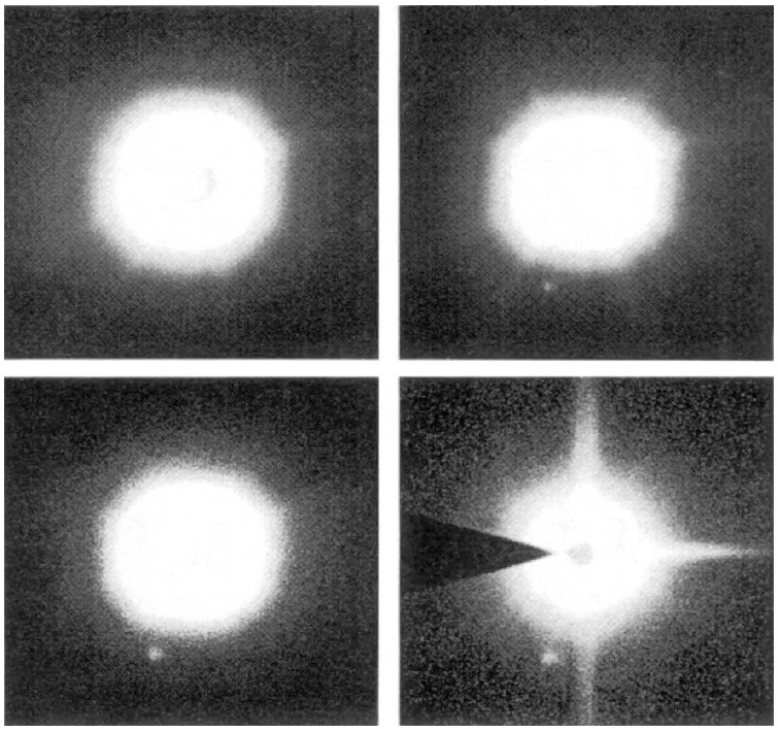
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Hubble Deep Field (1996)
342 frames – 4 filters – 140,8 h exposure time *Credit: R. Williams/STScI/NASA*

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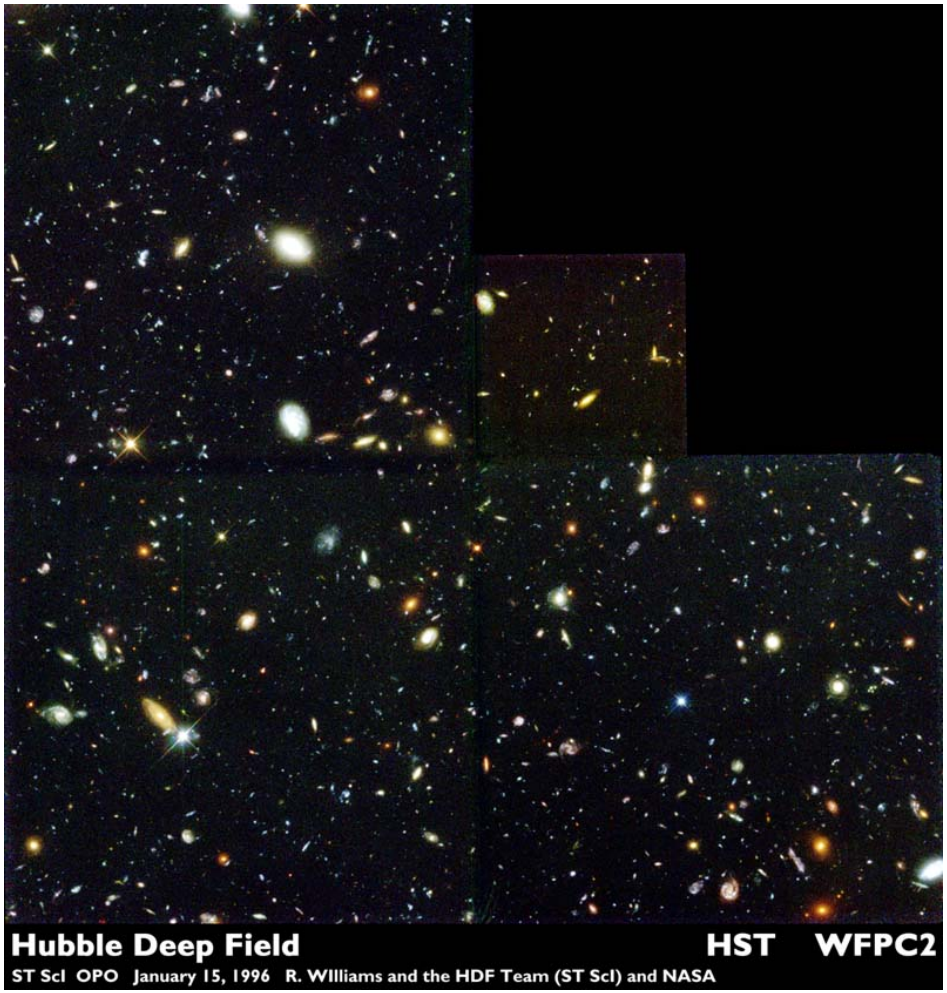


Discovery of brown dwarf GI 229 B

Nakajima et al. (1995), Nature

The CCD revolution in astronomy (2/2)

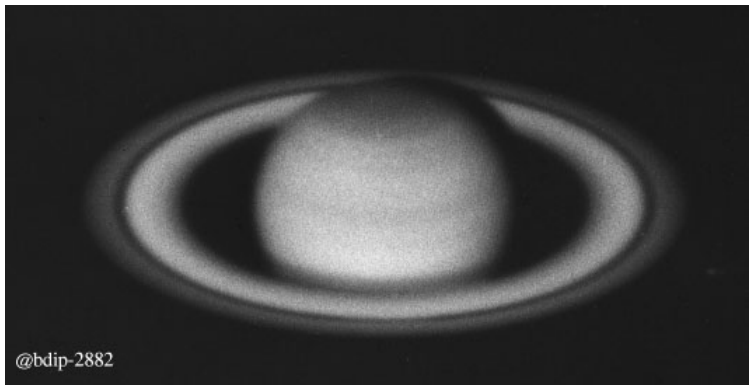
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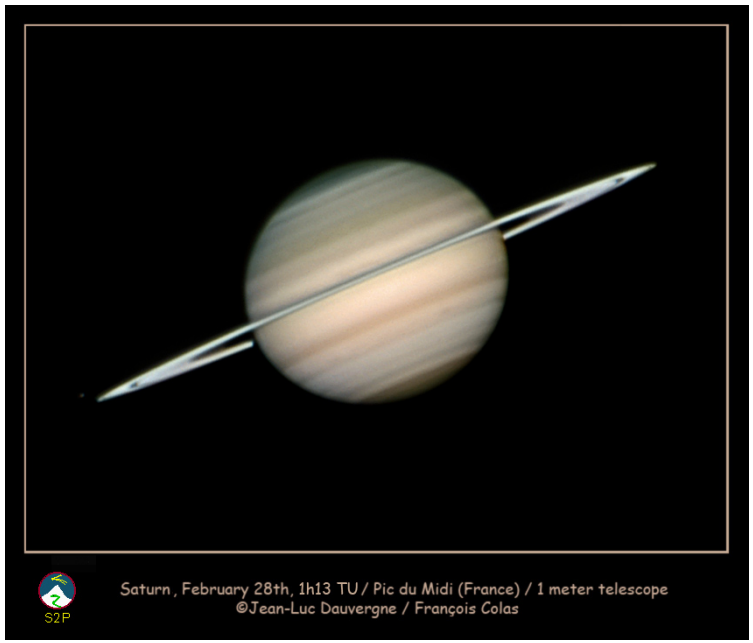
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28/02/1945



28/02/2009

Saturn at Pic du Midi T1M

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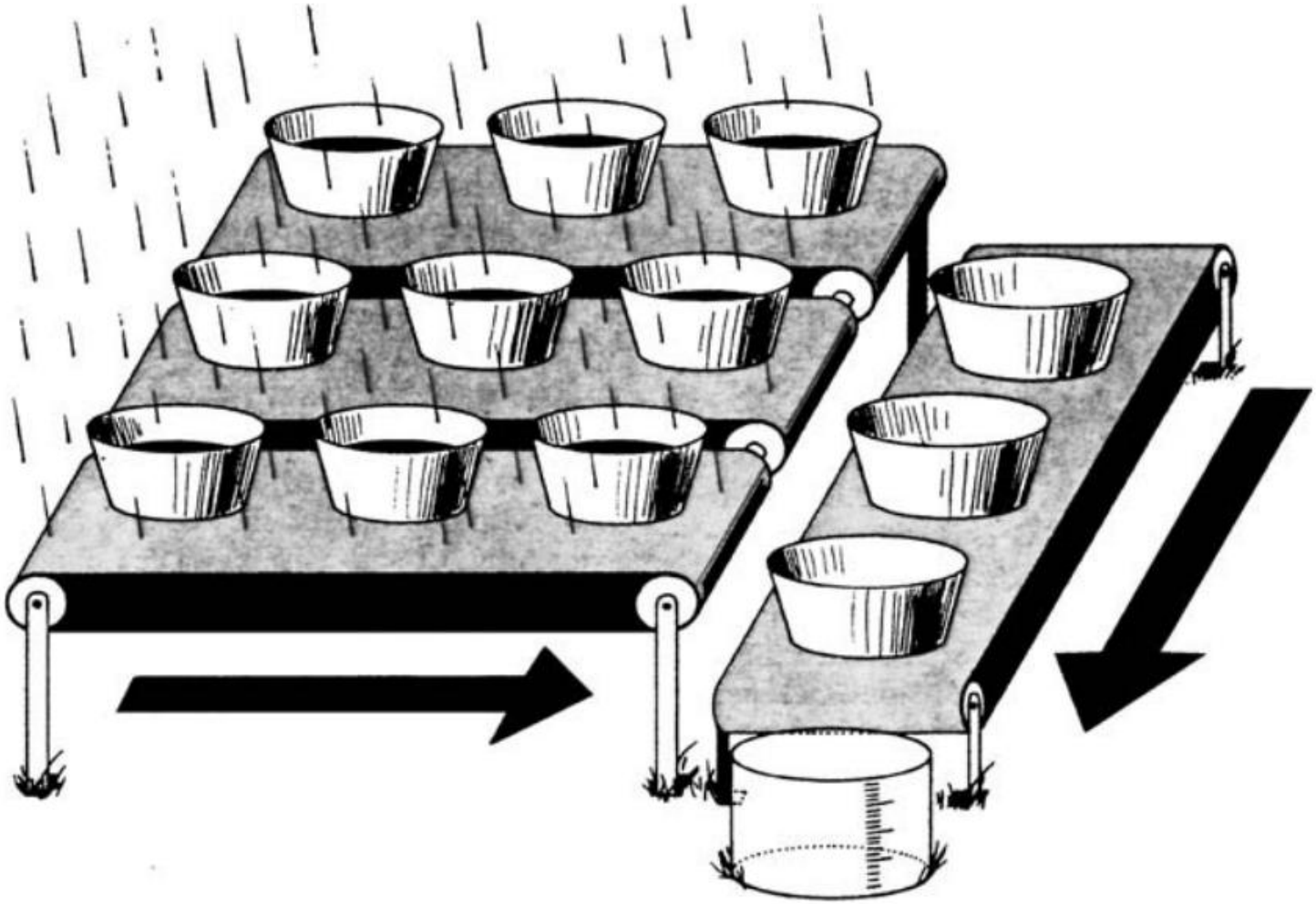
3 Structure and operation of a CCD
- Basic concept
- Photon detection
- Photoelectron storage
- Readout: charge transfer and digitization
- First analysis of a CCD image

4 Defects of CCD detectors and frame processing

5 CCD Cameras available at UM

6 Bibliographic ressources

Basic concept



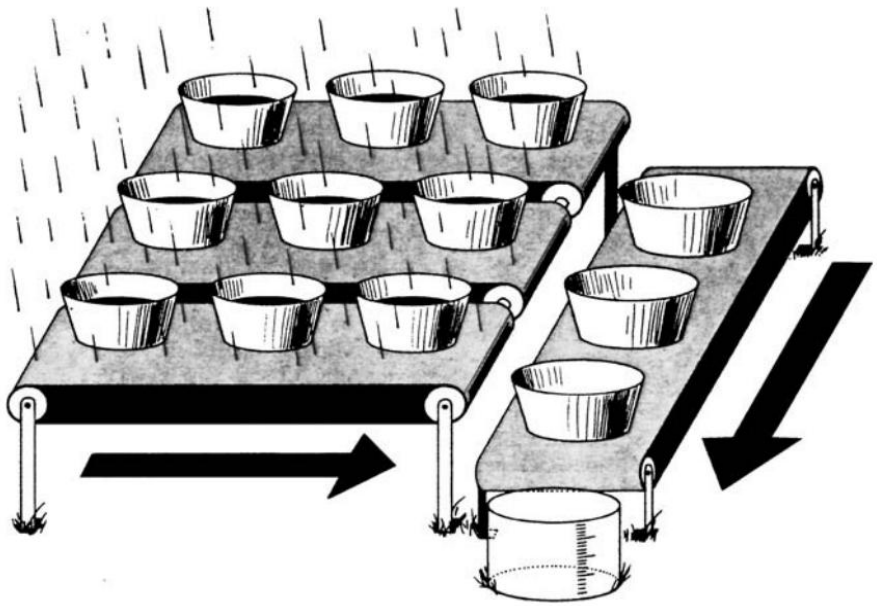
The water bucket analogy

Handbook of CCD Astronomy, from Janesick & Blouke (1987), Sky & Telescope

Basic concept

1. Photon detection
   - photoelectric effect
     - Explained by A. Einstein (1905)
     - Nobel prize 1921
   - 1 photon $\rightarrow$ 1 e^-
2. Photo-electrons are stored during the exposure
3. and then transfered to the readout register
4. Amplification/mesurement/analog-to-digital conversion

- Remark : recent digital cameras rely on a slighly different technology: APS-CMOS



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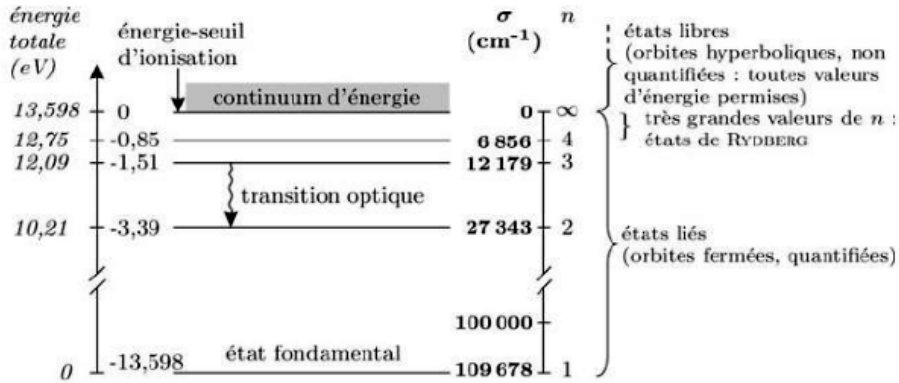
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`https://box.in2p3.fr/s/eMMQpkLtZmEm9Xy`

Credit: ASM / Observatoire de Paris

Photon detection

- Photoelectric effect
  - For $E_{\mathrm{phot}} > E_1$
    - Sillicium : $E_1 = 1.14$ eV
  - 1 photon $\rightarrow$ 1 e^-
- Material properties
  - Absorbs light efficiently
  - Emitted electrons are not re-absorbed
    - $\ell_{mfp\,e^-} > \ell_{mfp\,\gamma}$
  - Si: efficient for $E_{\mathrm{phot}} \sim 1.1 - 10s$ eV
    near-IR $\rightarrow$ soft X-rays
- Loss mechanisms
  - sound (phonon) production
  - charge diffusion



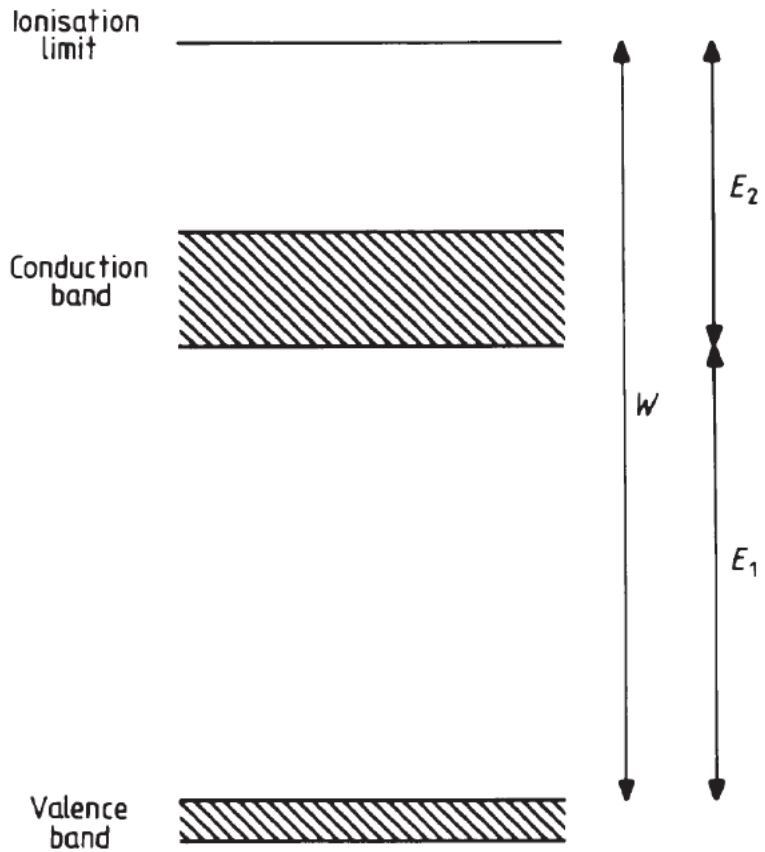
Energy diagram of hydrogen

Physique Atomique et Spectroscopie Optique

M. Moisan, D. Kéorack & L. Stafford

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Energy levels in a semi-conductor

Credit: R. Kitchin, Astrophysical Techniques

Photoelectron storage

- e^-/hole pair lifetime:

  $\tau \sim 100\ \mu$s
  - trapping in potential well

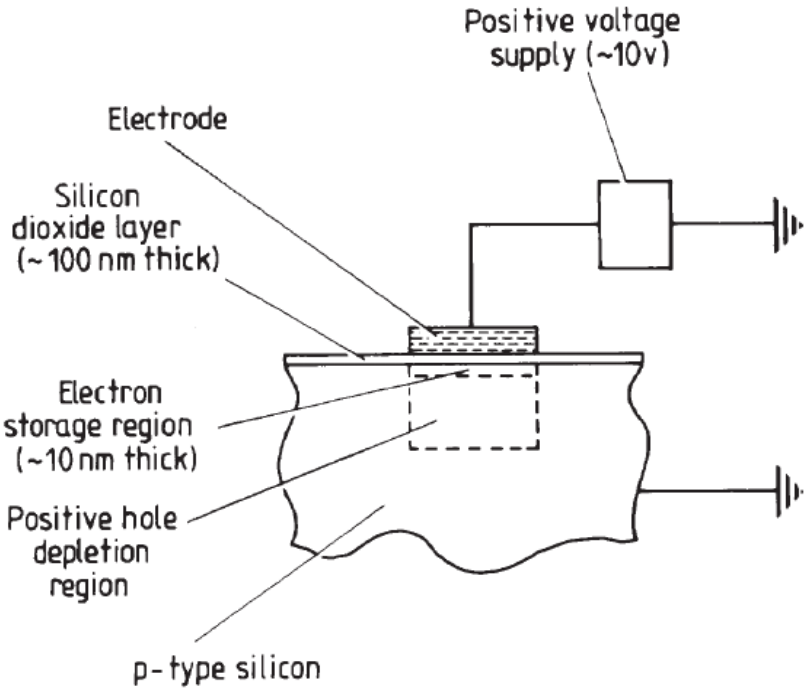
- MOS structure
  - hole depletion zone
  - e^- storage zone

- Arranged in 2-D array

➔ Pixel size
  - $2 - 30\ \mu$m

➔ Array format
  - $500 - 8\,000$ photo sensors

➔ Well capacity
  - $50\,000 - 100\,000 e^-$



Gate structure

Credit : R. Kitchin, Astrophysical Techniques

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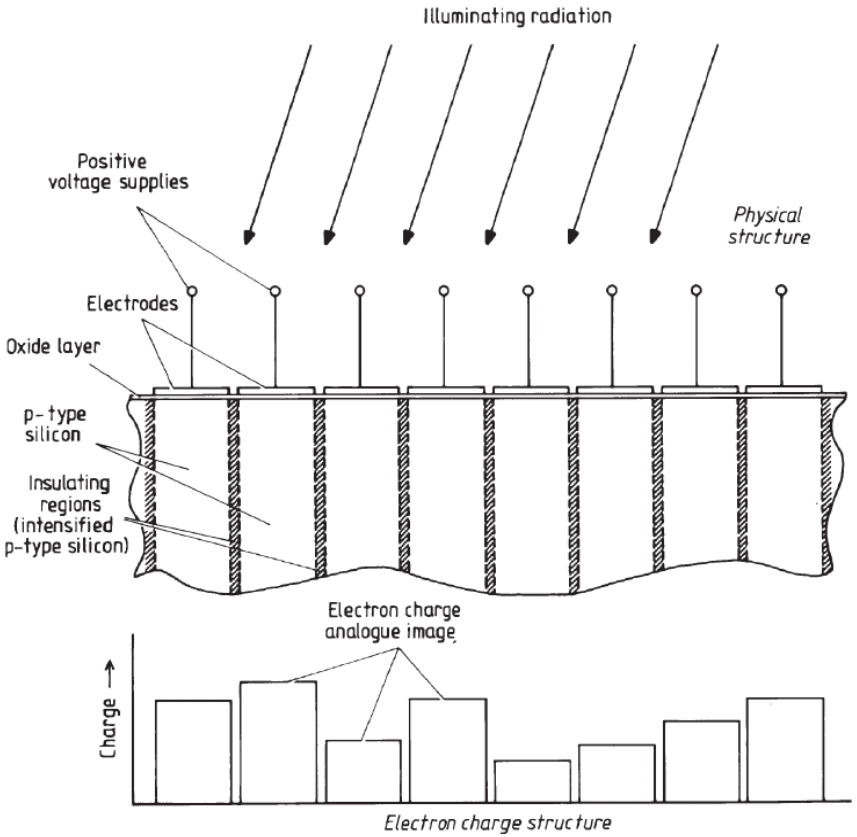
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Pixel arrangement along a column

Credit : R. Kitchin, Astrophysical Techniques

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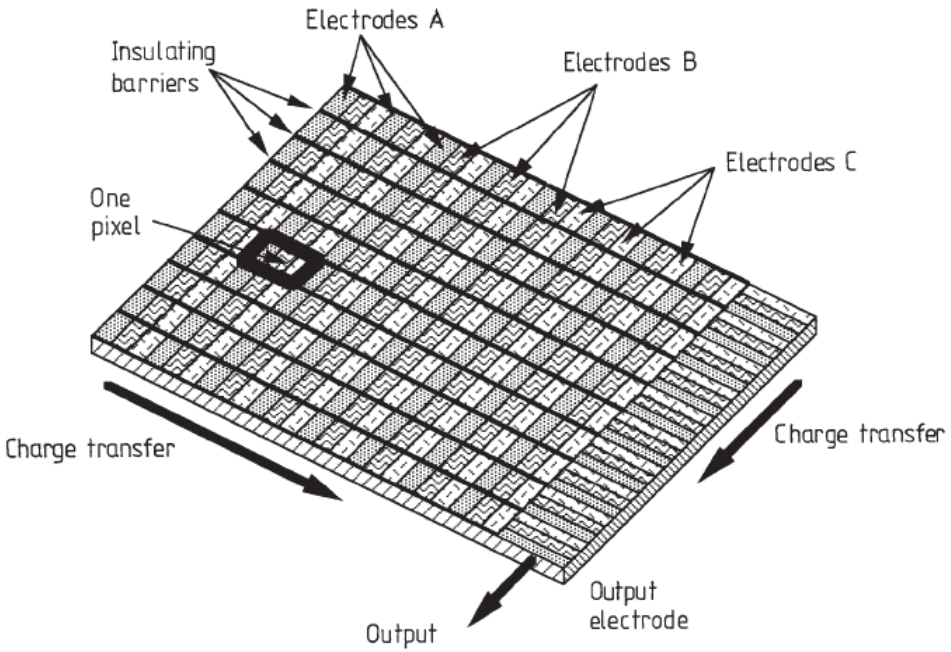
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Structure of a CCD array

Credit : R. Kitchin, Astrophysical Techniques

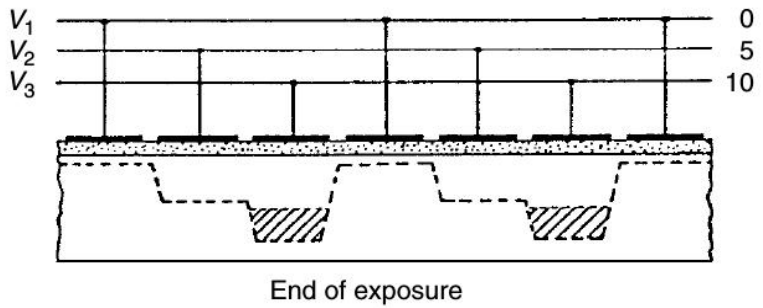
Readout: charge transfer and digitization

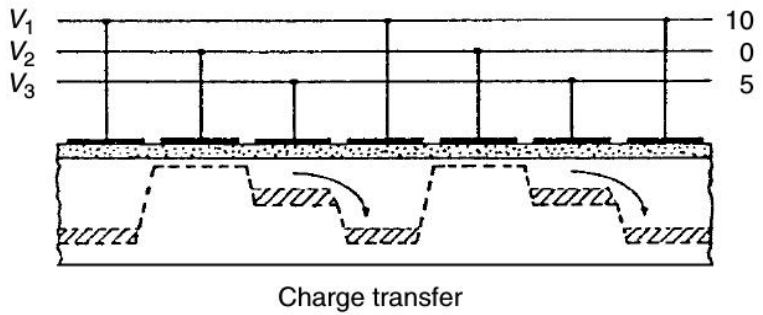
- Transfer charges along rows
  - Clock gates potentials
- Imperfect transfer

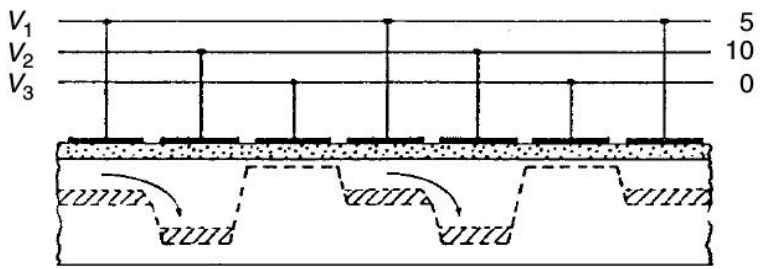
➔ CTE: transfer efficiency
➔ CTI: transfer inefficiency
  - CTI$\sim 10^{-5} - 10^{-6}$

- A/D converter
  - voltage ➔ integers
  - "unit": DN or ADU

➔ (inverse) Gain
  - $1 - 10\ e^-$/ADU







Voltage operation in a 3-phase CCD

Handbook of CCD Astronomy, from Walker (1987)

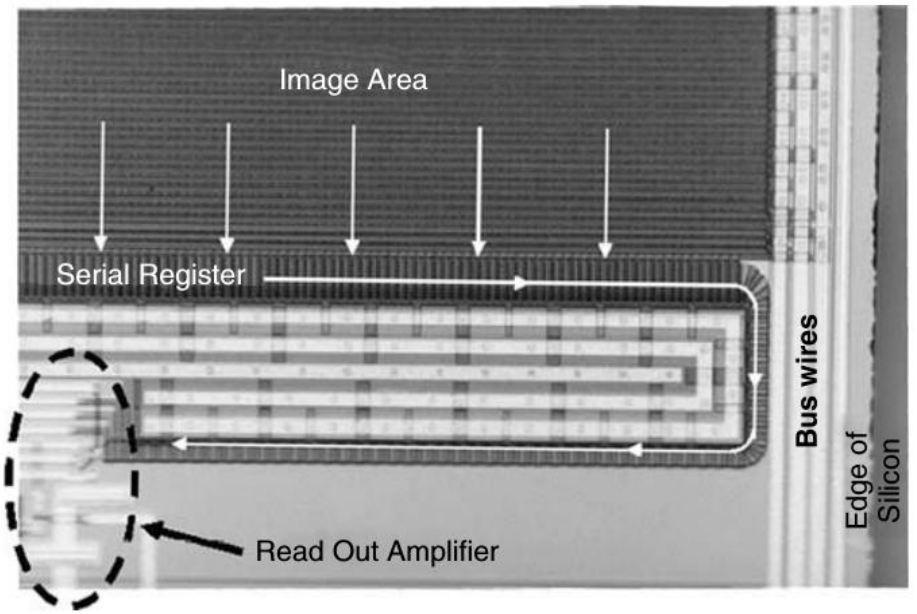
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Microphotography of a CCD

Handbook of CCD Astronomy

First analysis of a CCD image

- Image $\longleftrightarrow$ integer array
  - in ADU, related to the illumination at each CCD photo sensor
- 1 CCD photo sensor ➔ 1 image pixel
- 16 bits : 2^{16} possible values
  - min : $[0000000000000000]_2 = 0$

➥ : black pixel

  - max : $[1111111111111111]_2 = 2^{16} - 1 = 65\,535$

➥ white pixel

- Above, the pixel saturates
  - readout : blooming



Open cluster NGC 869 observed with T120 at Observatoire de Haute-Provence

Credit: Master Astrophysique / Cosmos Champs Particules Université de Montpellier

First analysis of a CCD image

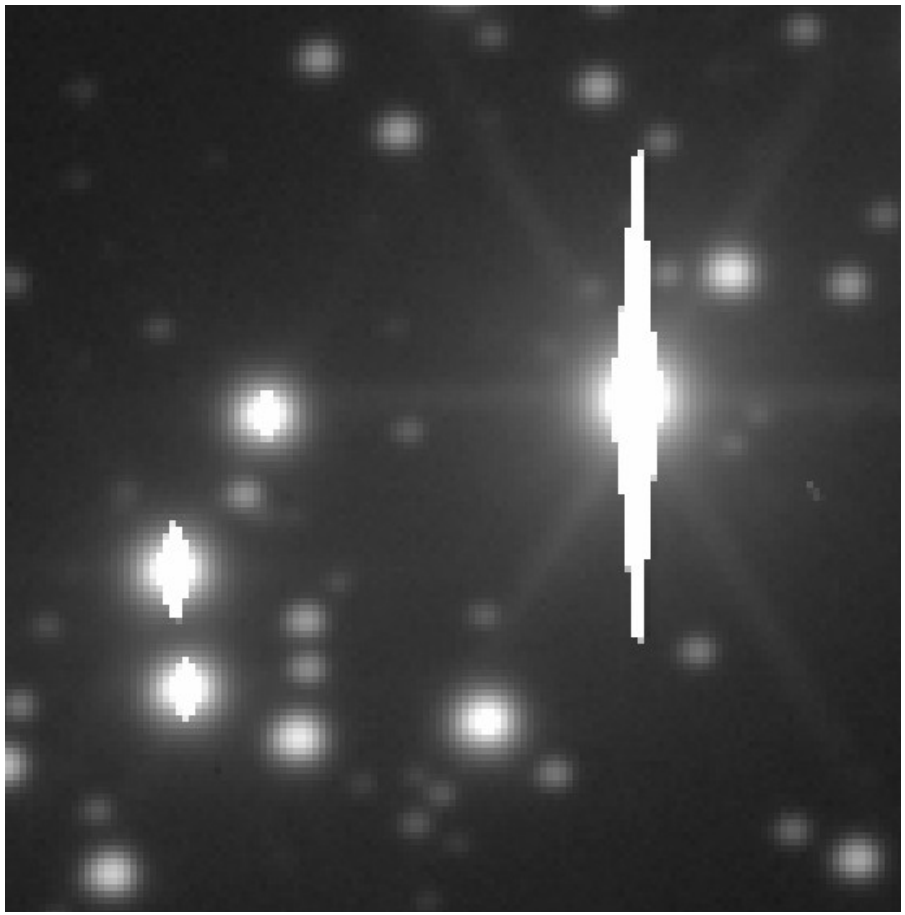
- Image $\longleftrightarrow$ integer array
  - in ADU, related to the illumination at each CCD photo sensor
- 1 CCD photo sensor ➔ 1 image pixel
- 16 bits : 2^{16} possible values
  - min : $[0000000000000000]_2 = 0$

➥ : black pixel

  - max : $[1111111111111111]_2 = 2^{16} - 1 = 65\,535$

➥ white pixel

- Above, the pixel saturates
  - readout : blooming



Open cluster NGC 869 observed with T120 at Observatoire de Haute-Provence

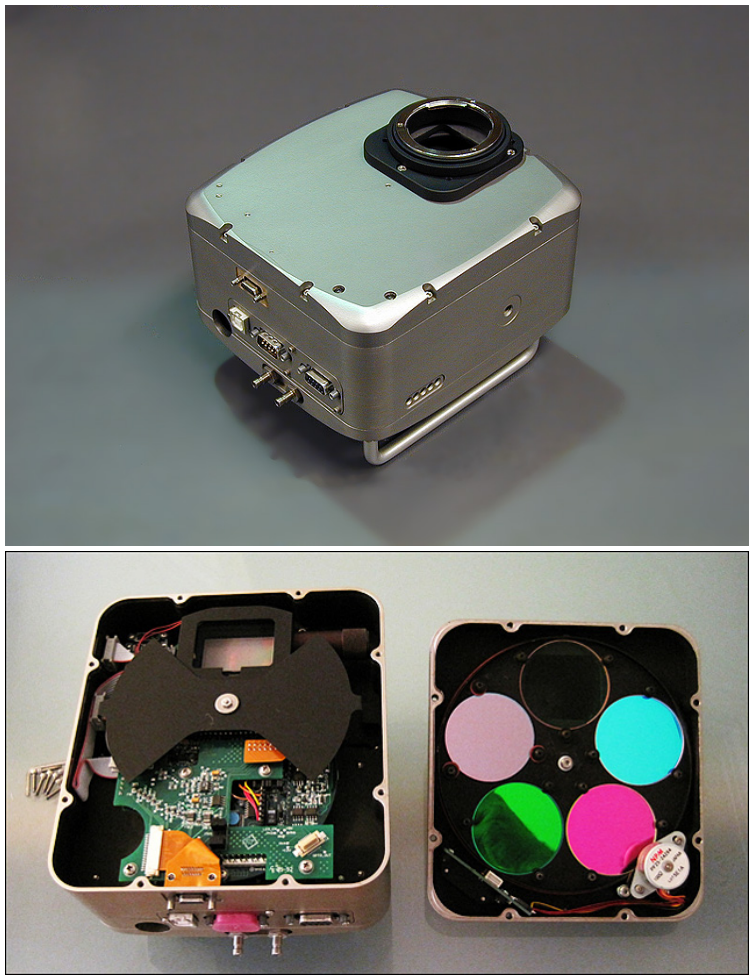
Credit: Master Astrophysique / Cosmos Champs Particules Université de Montpellier

Obtaining a color CCD image

- Each photo sensor is only sensitive to illumination

➥ black and white image

- Reconstruction of a color image using several filters
  - possible to produce "false color" images
- A digital camera uses a color filter array (CFA)



SBIG STL-11000M and filtetr wheel

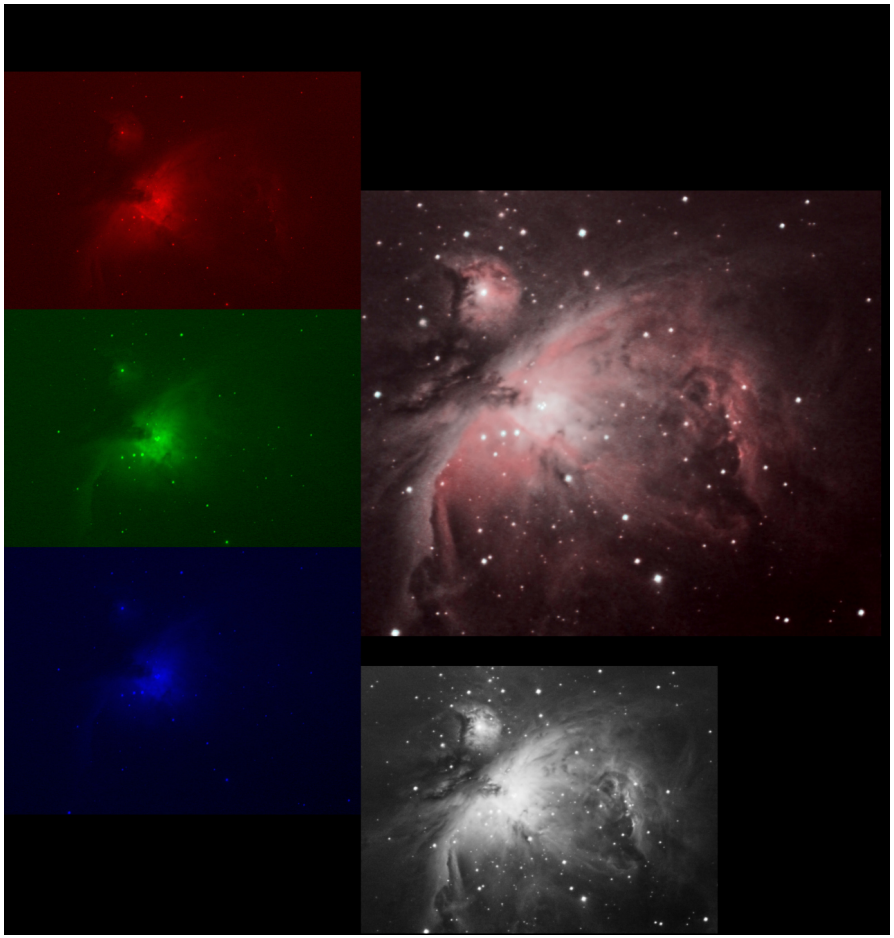
Credit: Santa Barbara Instrument Group

Obtaining a color CCD image

- Each photo sensor is only sensitive to illumination

➥ black and white image

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  - possible to produce "false color" images
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Nebula M42 observed with the Eichens 16 cm refractor at Observatoire Astronomique de la FdS

Credit: J. Morin/FdS Montpellier

Obtaining a color CCD image

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➥ black and white image

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Nebula M16 observed by the Hubble space telescope
$R/V/B \longleftrightarrow [S\text{II}]/H\alpha+[N\text{II}]/[O\text{III}]$

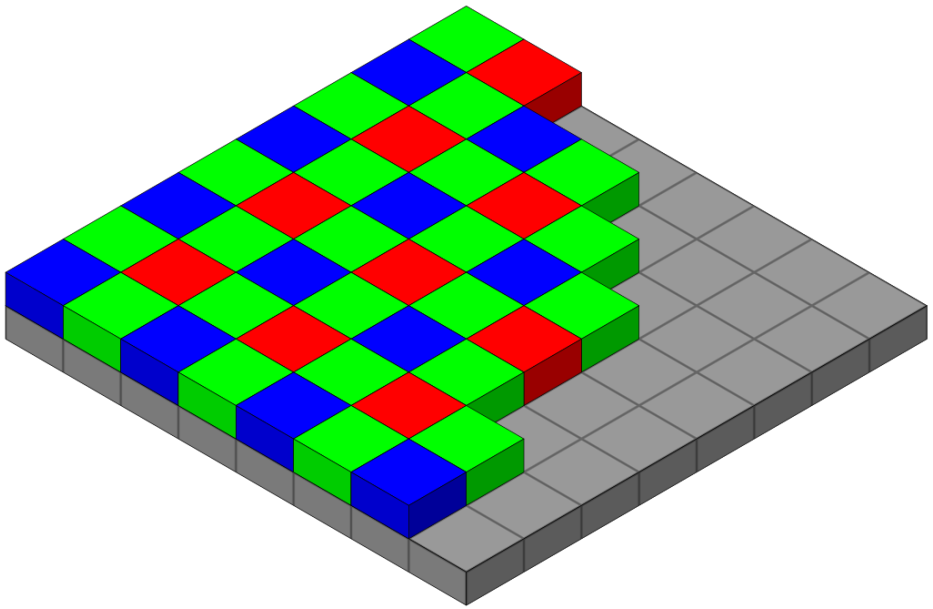
Credit: NASA, ESA and the Hubble Heritage Team (STScI/AURA)

Obtaining a color CCD image

- Each photo sensor is only sensitive to illumination

➥ black and white image

- Reconstruction of a color image using several filters
  - possible to produce “false color” images
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CFA: Bayer layout

Source: Wikimedia Commons

Outline

1 Enseignement d'astrophysique en Master Astrophysique/CCP

2 The CCD revolution in astronomy

3 Structure and operation of a CCD

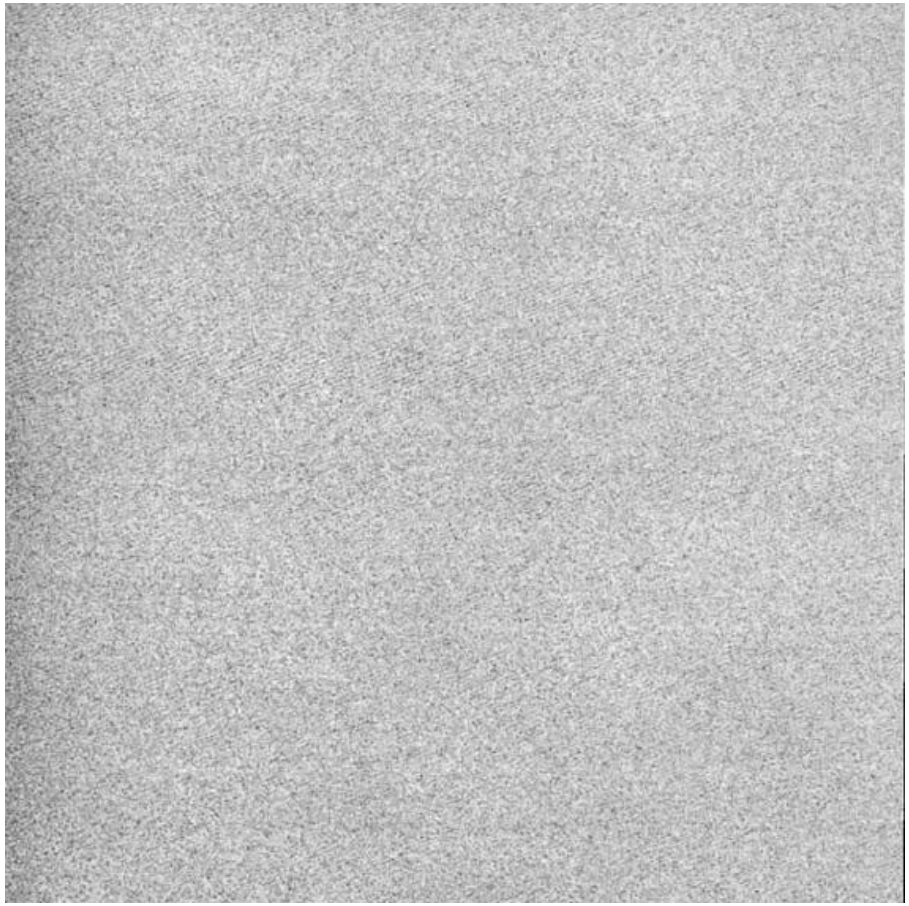
4 Defects of CCD detectors and frame processing

5 CCD Cameras available at UM

6 Bibliographic ressources

Defects of CCD detectors

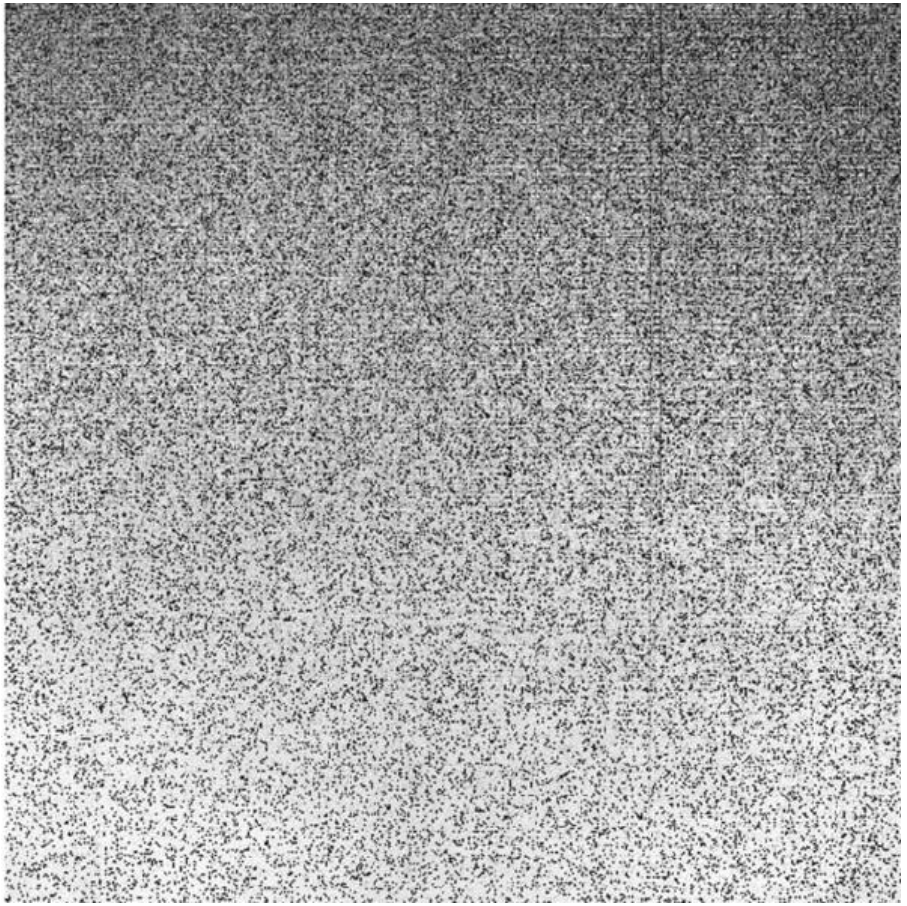
- CCD readout
  - Bias/Offset frame
  - Readout noise
- CCD thermal noise
  - Dark frame
  - Dark current
- CCD + optics field inhomogeneities
  - Flat field frame



Bias frame, credit: Handbook of CCD Astronomy

Defects of CCD detectors

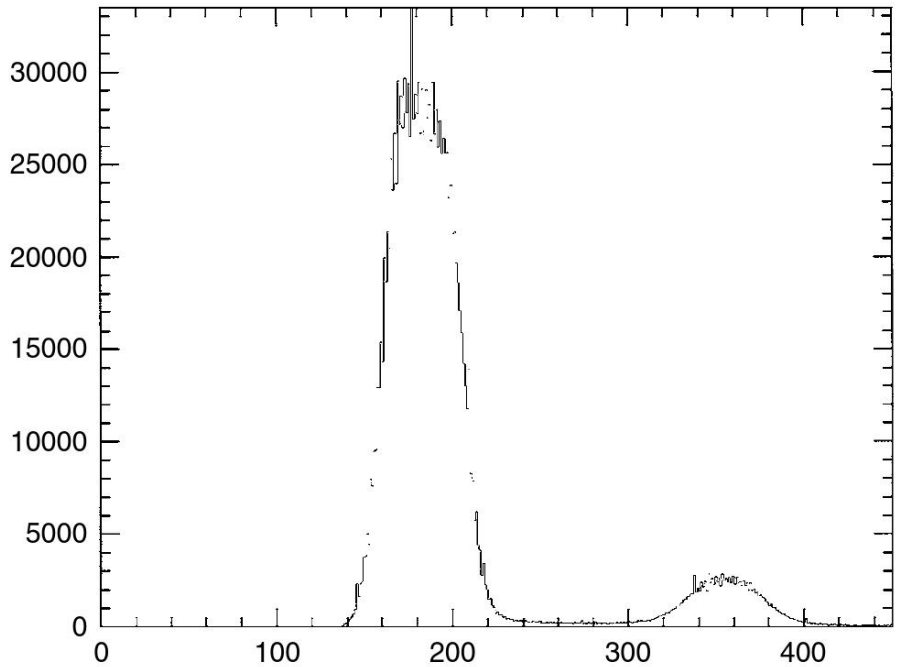
- CCD readout
  - Bias/Offset frame
  - Readout noise
- CCD thermal noise
  - Dark frame
  - Dark current
- CCD + optics field inhomogeneities
  - Flat field frame



Dark frame, credit: Handbook of CCD Astronomy

Defects of CCD detectors

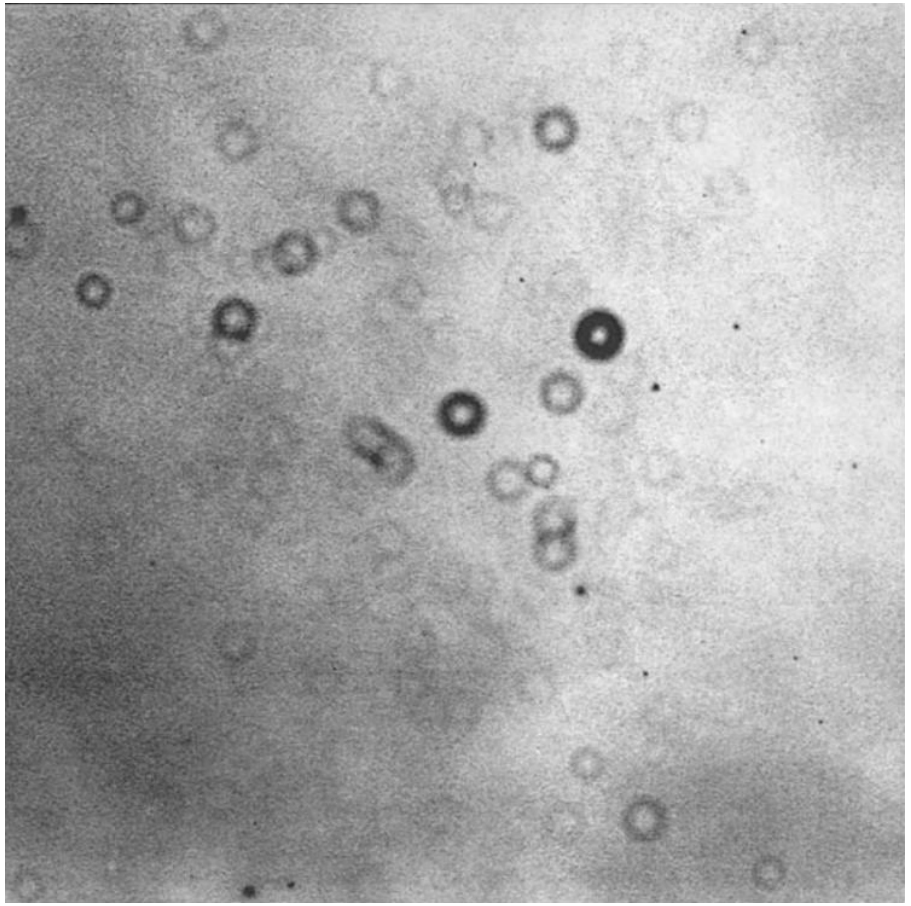
- CCD readout
  - Bias/Offset frame
  - Readout noise
- CCD thermal noise
  - Dark frame
  - Dark current
- CCD + optics field inhomogeneities
  - Flat field frame



Dark frame histogram, credit: Handbook of CCD Astronomy

Defects of CCD detectors

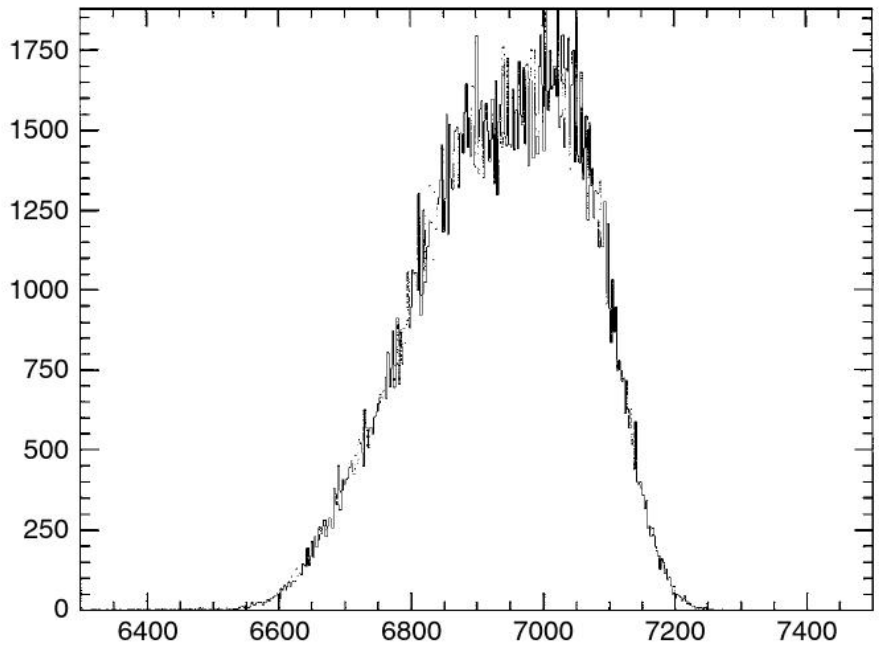
- CCD readout
  - Bias/Offset frame
  - Readout noise
- CCD thermal noise
  - Dark frame
  - Dark current
- CCD + optics field inhomogeneities
  - Flat field frame



Flat field frame, credit: Handbook of CCD Astronomy

Defects of CCD detectors

- CCD readout
  - Bias/Offset frame
  - Readout noise
- CCD thermal noise
  - Dark frame
  - Dark current
- CCD + optics field inhomogeneities
  - Flat field frame



Flat field histogram, credit: Handbook of CCD Astronomy

Image processing and CCD characterization

- Final image :

$$S_{ij} = \frac{O_{ij} - \overline{D}_{ij}(t_{\exp}, T)}{\overline{F}_{ij}}$$

- Signal to noise ratio:

$$\frac{\mathrm{S}}{\mathrm{N}} = \frac{N_*}{\sqrt{N_* + n_{\mathrm{pix}}(N_S + N_D + N_R^2)}}$$

- CCD characterization

$$\mathrm{Gain} = \frac{(\bar{F}_1 + \bar{F}_2) - (\bar{B}_1 + \bar{B}_2)}{\sigma^2_{F_1 - F_2} - \sigma^2_{B_1 - B_2}}, \qquad \text{Read noise} = \frac{\mathrm{Gain} \cdot \sigma_{B_1 - B_2}}{\sqrt{2}}.$$

➔ valid for imaging, things are a bit more complicated for spectroscopy!

Outline

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5 CCD Cameras available at UM
- SBIG STL-11000M CDD Camera
- Atik 460EX CCD Camera
- IRiS: FLI Proline 4240

6 Bibliographic ressources

SBIG STL-11000M CDD Camera

- SBIG STL-11000M
- KAI-11002M



SBIG STL-11000M

SBIG STL-11000M CDD Camera

- SBIG STL-11000M
- KAI-11002M

STL-11000M/CM Typical Specifications	
Imaging CCD	Kodak Enhanced KAI-11002M/CM, Class 1 or Class 2
Column Defects	Class 1 has no column defects, Class 2 allows up to 2 column defects
Pixel Array	4008 x 2672 pixels, 36 x 24.7 mm
Total Pixels	11 million
Pixel Size	9 x 9 microns
Full Well Capacity (NABG)	50,000 e-
Dark Current	0.5 e-/pixel/second at 0 degrees C
Antiblooming	ABG only
Shutter	Electromechanical and Electronic
Exposure	0.001 to 3600 seconds, 10ms resolution
Correlated Double Sampling	Yes
A/D Converter	16 bits
A/D Gain	0.8e^-/ADU unbinned, 1.6e- binned
Read Noise	13e^- RMS
Binning Modes	1 x 1, 2 x 2, 3 x 3
Full Frame Download	26 seconds
Cooling - standard	Two-Stage Thermoelectric, Water Assist, -40 C from Ambient Typical
Temperature Regulation	±0.1°C
Power	10 - 18VDC, 12VDC nominal, Universal AC to 12VDC desktop supply
Computer Interface	USB 1.1
Computer Compatibility	Win 98 / NT / 2000 / Me / XP / Mac OS-X / Vista 32 bit / Linux (3rd pty)
Guiding	Dual CCD Self-Guiding Standard, Remote Guiding Head Optional
Dimensions	6.5 x 6 x 3.5" (16.5 x 15.2x8.9cm)
Weight	4 pounds (1.8 Kg) without filters
Internal Filter Carousel	5 positions for 48mm threaded cells or 2" unmounted filters (optional)
Mounting	2" nosepiece included
Backfocus	Approximately 1.7 inches (~4.3 cm) with 2" nosepiece attached

SBIG STL-11000M specifications

SBIG STL-11000M CDD Camera

- SBIG STL-11000M
- KAI-11002M

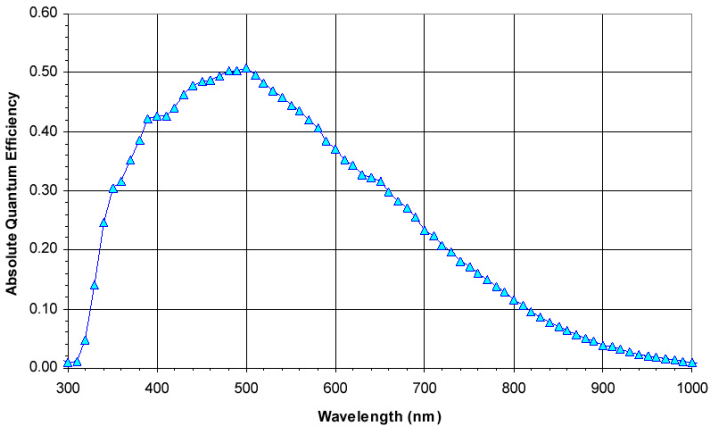


Figure 11 - Monochrome with Microlens Quantum Efficiency

SBIG STL-11000M QE

Atik 460EX CCD Camera

- Atik 460EX
- Sony ICX694



Atik 460EX

Atik 460EX CCD Camera

- Atik 460EX
- Sony ICX694

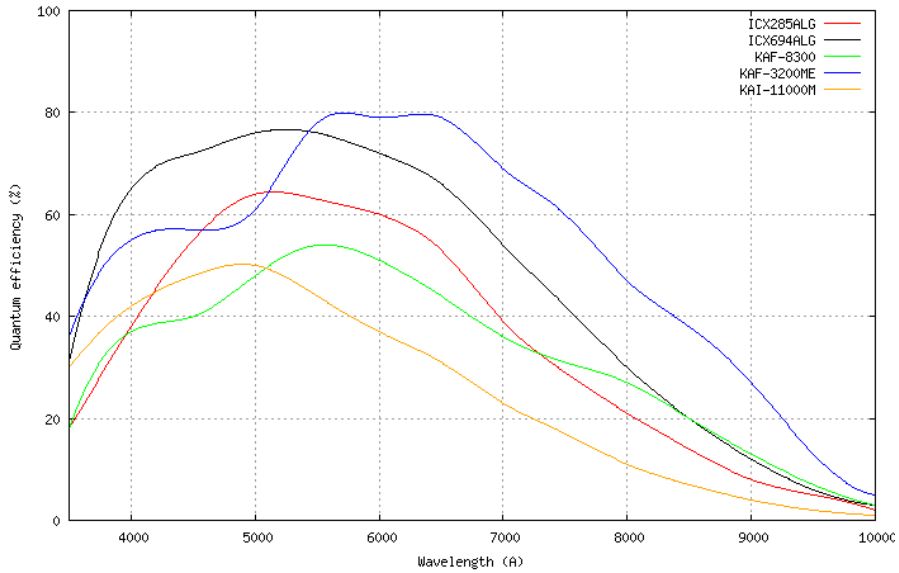
Technical Specification

- Sensor Type: CCD - Sony ICX694 /5
- Horizontal Resolution: 2750 pixels
- Vertical Resolution: 2200 pixels
- Pixel Size: 4.54 µm x 4.54 µm
- ADC: 16 bit
- Readout Noise: 5e- typical value
- Gain Factor: 0.27e-/ADU
- Full Well: ~18,000 e-
- Dark Current: ~0.0004 electrons/second at -10°
- Interface: Mini-USB 2.0 High Speed
- Power: 12v DC 1A
- Maximum Exposure Length: Unlimited
- Minimum Exposure Length: 1/1000 s
- Cooling: Termoelectric set point with max ΔT=-25°C
- Weight: approx. 400 gr
- Backfocus: 13mm
- Available as mono or colour

Atik 460EX specifications

Atik 460EX CCD Camera

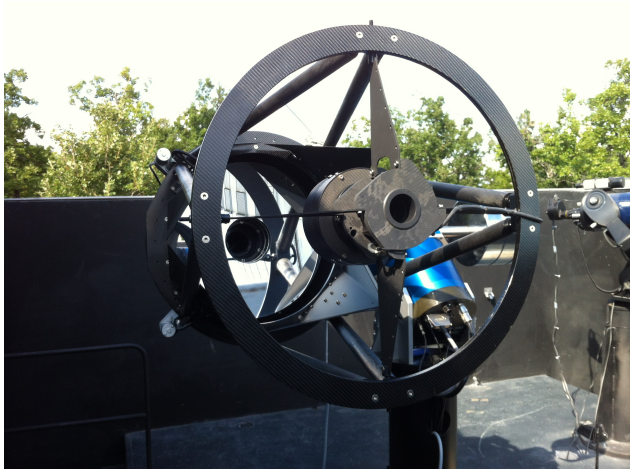
- Atik 460EX
- Sony ICX694



CCDs relative QEs, credit: Ch. Buil

IRiS: FLI Proline 4240

- FLI Proline 4240
- e2v CCD42-40-1-368
- `http://iris.lam.fr`



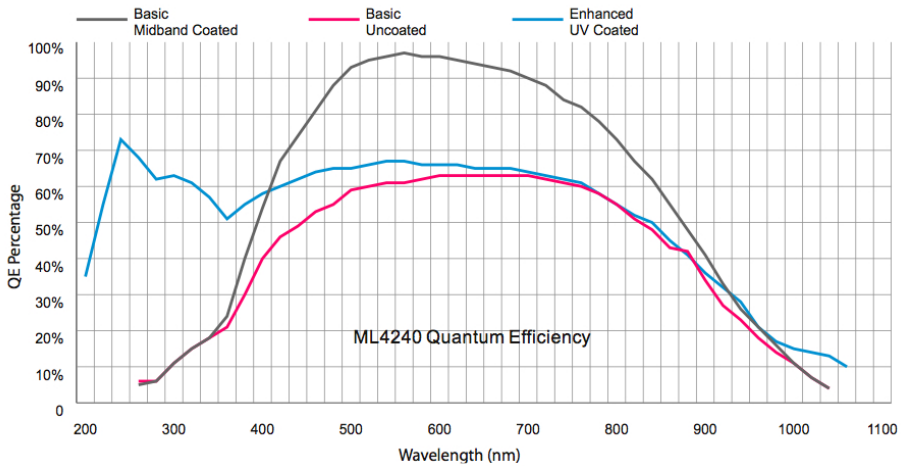
IRiS telescope



FLI Proline 4240

IRiS: FLI Proline 4240

- FLI Proline 4240
- e2v CCD42-40-1-368
- `http://iris.lam.fr`



Atik 460EX specifications

IRiS: FLI Proline 4240

- FLI Proline 4240
- e2v CCD42-40-1-368
- `http://iris.lam.fr`

Technical Data	
Sensor Type	Back Illuminated CCD
Sensor	e2v CCD42-40-1-368 (MB) or -153 (BB)
Active Pixels	2048 x 2048
Pixel Size (microns)	13.5 x 13.5 µm
Imaging Area (Diagonal)	27.6 X 27.6 mm (39 mm)
Full Well Capacity	100000 electrons
Typical_Readout Noise	10 e- RMS @ 500 kHz; 14 e- @ 2 MHz
Typical Gain	1.35e-/ADU
Dynamic Range	79.7 dB
Anti-Blooming	None
Cooling Method	Air (Optional liquid)
Max. Cooling (Air)	60°C below ambient
Temperature Stability	0.1°C
Dark Current (typical)	0.2 eps at -30C
Interface	USB 2.0
Digitization Clock	500 kHz and 2 MHz (software
Data Bit Depth	16 bits
Non-Linearity	<1%
Channels	1
Shutter	65 mm

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6 Bibliographic ressources

Bibliographic ressources

- *"Astronomie Astrophysique, Introduction, 5e édition"*, A. Acker, 2013, Dunod
  ISBN-13: 978-2100576401
- *"Essential Astrophysics"*, K.R. Lang, 2013, Springer
  ISBN-13: 978-3-642-35962-0
- *"L'Observation en Astrophysique"*, P. Léna et al., 2008, EDP Sciences
  ISBN-13: 978-2868838773
- *"Astrophysical Techniques"*, C.R. Kitchin, 2013, CRC Press
  ISBN-13: 978-1466511156
- Further readings
  - *"Handbook of CCD Astronomy"*, S.B. Howell,
    Cambridge Observing Handbooks for Research Astronomers,
    Cambridge University Press, 2006
    ISBN-13: 978-0521617628